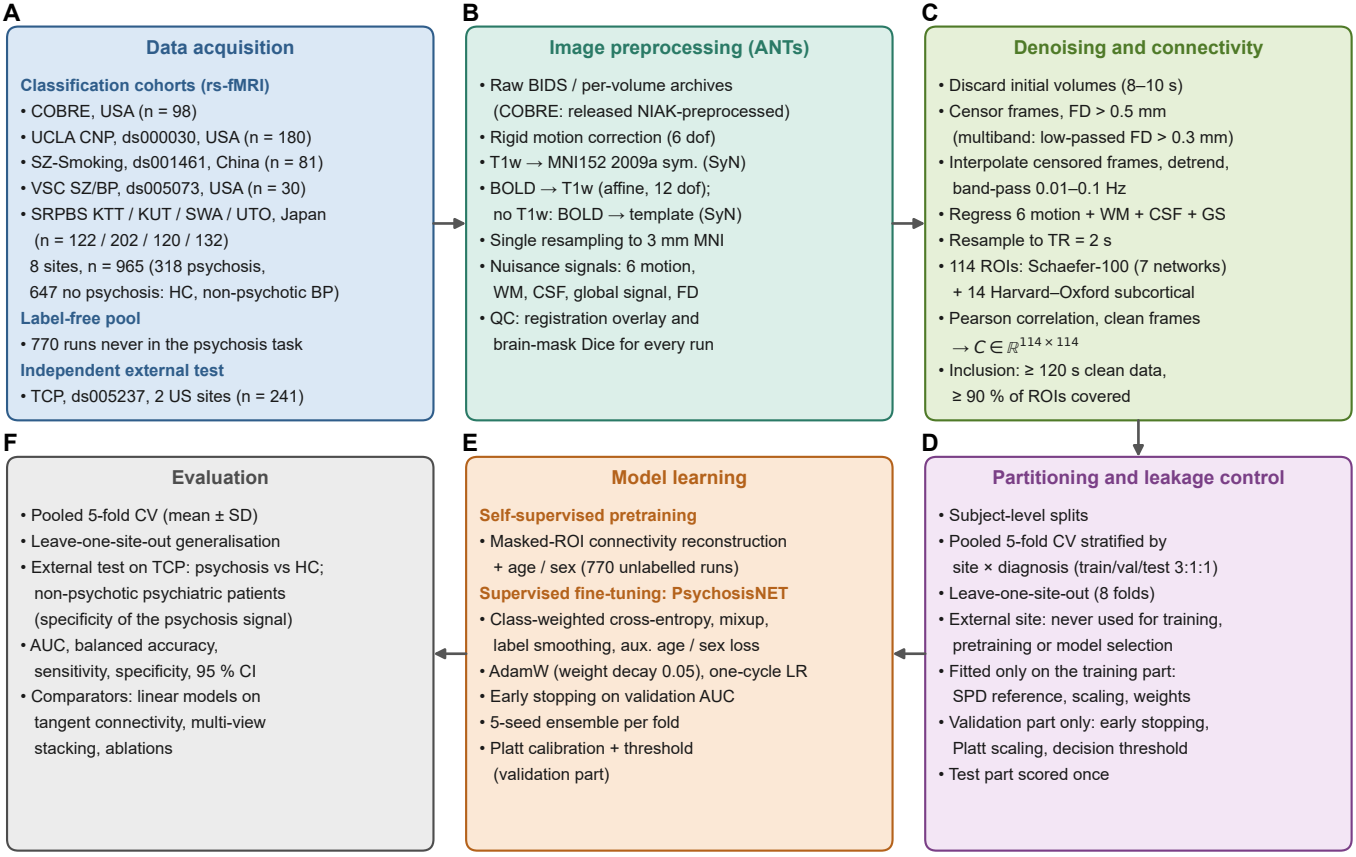

A
Data acquisition
Classification cohorts (rs-fMRI)
• COBRE, USA (n = 98)
• UCLA CNP, ds000030, USA (n = 180)
• SZ-Smoking, ds001461, China (n = 81)
• VSC SZ/BP, ds005073, USA (n = 30)
• SRPBS KTT / KUT / SWA / UTO, Japan
(n = 122 / 202 / 120 / 132)
8 sites, n = 965 (318 psychosis,
647 no psychosis: HC, non-psychotic BP)
Label-free pool
• 770 runs never in the psychosis task
Independent external test
• TCP, ds005237, 2 US sites (n = 241)
B
Image preprocessing (ANTs)
• Raw BIDS / per-volume archives
(COBRE: released NIAK-preprocessed)
• Rigid motion correction (6 dof)
• T1w → MNI152 2009a sym. (SyN)
• BOLD → T1w (affine, 12 dof);
no T1w: BOLD → template (SyN)
• Single resampling to 3 mm MNI
• Nuisance signals: 6 motion,
WM, CSF, global signal, FD
• QC: registration overlay and
brain-mask Dice for every run
C
Denoising and connectivity
• Discard initial volumes (8–10 s)
• Censor frames, FD > 0.5 mm
(multiband: low-passed FD > 0.3 mm)
• Interpolate censored frames, detrend,
band-pass 0.01–0.1 Hz
• Regress 6 motion + WM + CSF + GS
• Resample to TR = 2 s
• 114 ROIs: Schaefer-100 (7 networks)
+ 14 Harvard–Oxford subcortical
• Pearson correlation, clean frames
→ $C \in \mathbb{R}^{114 \times 114}$
• Inclusion: ≥ 120 s clean data,
≥ 90 % of ROIs covered
D
Partitioning and leakage control
• Subject-level splits
• Pooled 5-fold CV stratified by
site × diagnosis (train/val/test 3:1:1)
• Leave-one-site-out (8 folds)
• External site: never used for training,
pretraining or model selection
• Fitted only on the training part:
SPD reference, scaling, weights
• Validation part only: early stopping,
Platt scaling, decision threshold
• Test part scored once
E
Model learning
Self-supervised pretraining
• Masked-ROI connectivity reconstruction
+ age / sex (770 unlabelled runs)
Supervised fine-tuning: PsychosisNET
• Class-weighted cross-entropy, mixup,
label smoothing, aux. age / sex loss
• AdamW (weight decay 0.05), one-cycle LR
• Early stopping on validation AUC
• 5-seed ensemble per fold
• Platt calibration + threshold
(validation part)
F
Evaluation
• Pooled 5-fold CV (mean ± SD)
• Leave-one-site-out generalisation
• External test on TCP: psychosis vs HC;
non-psychotic psychiatric patients
(specificity of the psychosis signal)
• AUC, balanced accuracy,
sensitivity, specificity, 95 % CI
• Comparators: linear models on
tangent connectivity, multi-view
stacking, ablations